



CCS6314

CRYPTOGRAPHY AND DATA SECURITY

(DR. CHAN WAI KOK)

NOTE: SLIDES TAKEN FROM DR TEA (FROM PREVIOUS SEMESTER) & MODIFIED ACCORDINGLY

Syllabus Contents:

- What will you learn in this course?
 1. Introduction to data and information security.
 2. Mathematical foundations.
 3. Classical and modern symmetric ciphers.
 4. Modern asymmetric ciphers.
 5. Various cryptographic algorithms for specific objective.
 6. Key management and distribution.
 7. Recent trends in / futures of cryptography.



Course Content:

- Solely **100%** coursework-based – **No final examination.**
- Throughout the trimester:
 1. **Quizzes:** 2 (**Physical-based** with 10% each) – Total: **20%**
 2. **Tests:** 2 (**Physical-based** with 20% each) – Total: **40%**
 3. **Assignment:** 1 or 2 (TBD later) (**Group-based** with report and oral presentations)
 - Total: **40%**

Course Structures:

➤ Lecture Session (2 hours per week):

Lecture Group	Day	Time	Venue
TC1L	Thursday	6 pm to 8pm	CNMX 1004
TC2L	Friday	10am-12 am	AR1009-FCI

Tutorial Sessions (2 hours per week):

Tutorial Group	Day	Time	Venue
TT1L	Wednesday	12 – 2 pm	AR2005 Cisco
TT2L	Tuesday	8am – 10 am	AR2005 Cisco
TT3L	Friday	5pm-7 pm	AR2003
TT4L	Wednesday	4pm to 6 pm	AR2003
TT5L	Tuesday	4pm to 6pm	AR2003
TT6L	Monday	4pm to 6pm	AR2003

Main Reference Books

1. Stallings, W. (2020). Cryptography and Network Security: Principles and Practice (8th ed.). Pearson.
2. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
3. Stinson, D. R. & Paterson, M. (2019). Cryptography: Theory and Practice (4th ed.). Taylor & Francis.
4. Kahate, A. (2019). Cryptography and Network Security (4th ed.). McGraw-Hill.